Week 2 Lab: Principal Component Analysis (PCA)

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Course:BFOR516 - Adv Data Analytics for Cyber **Date:** February 6, 2026

Week 2 Lab

AI Usage Statement

This lab was completed with assistance from ****Google Antigravity AI****. The AI tool was used for

1. **Code Generation**: Generating boilerplate Python code for PCA implementation using scikit-learn.

2. **Visualization**: Creating matplotlib code for the 2D scatter plot with proper formatting.

3. **Feature Analysis**: Implementing code to extract and sort PCA loadings for feature contribution analysis.

4. **Documentation**: Adding comprehensive comments to explain each step of the PCA workflow.

5. **Debugging**: Fixing Unicode encoding errors and ensuring cross-platform compatibility.

Important Note: While AI assisted with code syntax and implementation details, the interpretation of results, analytical reasoning, and understanding of PCA concepts are my own work. The core logic, interpretation, and analytical approach were designed by Spencer Kone. AI was used as a coding assistant for syntax, libraries, and implementation details.

Introduction

This lab focuses on performing Principal Component Analysis (PCA) on a dataset of 134,198 tweets to reduce dimensionality while retaining significant information. The dataset contains 58 numeric features related to tweet content, user metrics, and linguistic patterns.

The objective is to:

1. Preprocess and standardize the data.

2. Apply PCA to reduce the feature space while retaining 90% of the variance.

3. Visualize the transformed data to identify patterns between fake and real news.

Dimensionality reduction is critical in cybersecurity and data analytics to improve model performance, reduce overfitting, and visualize high-dimensional data.

```
"""

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.decomposition import PCA

from sklearn.preprocessing import StandardScaler

STEP 1: LOAD AND PREPARE THE DATASET

print("=" * 80)

print("STEP 1: LOADING AND PREPARING THE DATASET")

print("=" * 80)

# Load the dataset

# AI assistance: Suggested using pd.read_csv with error handling

df = pd.read_csv('Features_For_Traditional_ML_Techniques.csv')

print(f"Dataset loaded successfully!")

print(f"Dataset shape: {df.shape}")

print(f"Columns: {df.columns.tolist()}\n")
```

```
# Prepare the data

# Remove non-numeric columns and target variable

# AI assistance: Provided the list of columns to drop based on the starter code

cols_to_drop = ['majority_target', 'statement', 'BinaryNumTarget', 'tweet', 'embeddings',
'following']

X_raw = df.drop(columns=cols_to_drop, errors='ignore')


# X_raw is the feature matrix used for PCA

# PCA is unsupervised, so it must not see the target (BinaryNumTarget)

# However, we keep y for visualization and interpretation later

y = df['BinaryNumTarget']


print(f"After dropping non-numeric and target columns:")

print(f"Feature matrix shape: {X_raw.shape}")

print(f"Number of features: {X_raw.shape[1]}")

print(f"Number of samples: {X_raw.shape[0]}\n")


# Handle any NaN values that might break the scaler

# AI assistance: Suggested using fillna(0) as a simple imputation strategy

X_raw = X_raw.fillna(0)

print(f"Missing values filled with 0")

print(f"Remaining NaN values: {X_raw.isna().sum().sum()}\n")
```

STEP 2: STANDARDIZE THE FEATURES

Standardize the dataset using StandardScaler

AI assistance: Provided the sklearn StandardScaler implementation

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_raw)
print(f"Features standardized using StandardScaler")
print(f"Scaled data shape: {X_scaled.shape}")
print(f"Mean of standardized features (should be ~0): {X_scaled.mean():.6f}")
print(f"Std of standardized features (should be ~1): {X_scaled.std():.6f}\n")
```

Student interpretation

Standardization is crucial before performing PCA for the following reasons:

1. SCALE SENSITIVITY: PCA is sensitive to the scale of features. Features with larger variances will dominate the principal components, even if they are not the most informative for the analysis.
2. VARIANCE EQUALITY: By standardizing (mean=0, std=1), we ensure all features contribute equally to the analysis based on their correlation structure, not their original measurement scales.
3. FAIR COMPARISON: In this dataset, different features may have vastly different ranges (e.g., word counts vs. sentiment scores). Standardization puts them on a level playing field.
4. NUMERICAL STABILITY: Standardization can improve the numerical stability of the PCA algorithm, leading to more reliable results.

STEP 3: APPLY PCA

Reduce dimensionality while retaining 90% of variance

AI assistance: Suggested using n_components=0.9 to automatically select components that explain 90% of variance

```
pca = PCA(n_components=0.9)

X_pca = pca.fit_transform(X_scaled)


print(f"PCA applied with 90% variance retention")

print(f"Transformed data shape: {X_pca.shape}")

print(f"Number of components kept: {pca.n_components_}")

print(f"Original number of features: {X_raw.shape[1]}")

print(f"Dimensionality reduction: {X_raw.shape[1]} -> {pca.n_components_}")

print(f"Compression ratio: {(1 - pca.n_components_/X_raw.shape[1])*100:.2f}%\n")
```

STEP 4: VISUALIZE THE PCA RESULTS

```
# Create a 2D scatter plot of the first two principal components
# AI assistance: Generated matplotlib code for creating the scatter plot with
# proper labels, legend, and color mapping
plt.figure(figsize=(10, 8))

# Color points by target variable (0: Fake, 1: Real)
colors = ['red', 'blue']
labels = ['Fake News', 'Real News']

for target, color, label in zip([0, 1], colors, labels):
    mask = y == target
    plt.scatter(X_pca[mask, 0], X_pca[mask, 1],
                c=color, label=label, alpha=0.6, s=30)

plt.xlabel(f'PC1 ({pca.explained_variance_ratio_[0]*100:.2f}% variance)', fontsize=12)
plt.ylabel(f'PC2 ({pca.explained_variance_ratio_[1]*100:.2f}% variance)', fontsize=12)
plt.title('PCA: First Two Principal Components\nFake vs Real News Classification',
          fontsize=14, fontweight='bold')
plt.legend(loc='best', fontsize=10)
plt.grid(True, alpha=0.3)

# Save the plot
# AI assistance: Suggested using bbox_inches='tight' and dpi=300 for better quality
plt.savefig('truth_seeker.png', dpi=300, bbox_inches='tight')
print("Plot saved as 'truth_seeker.png'")
plt.close()

print("Visualization complete!\n")
```

STEP 5: ANALYZE EXPLAINED VARIANCE

```
# Print number of components kept
print(f"Number of components kept: {pca.n_components_}\n")

# Print explained variance ratio of each component
print("Explained Variance Ratio by Component:")
print("-" * 80)
for i, var_ratio in enumerate(pca.explained_variance_ratio_, 1):
    print(f"  PC{i}: {var_ratio*100:.4f}% of variance")

# Compute and print total variance captured
total_variance = np.sum(pca.explained_variance_ratio_)
print(f"\nTotal variance captured: {total_variance*100:.4f}%")
```

Analysis question: Did PCA reduce the data effectively?

My interpretation: This analysis is my own

PCA reduced the dataset from {X_raw.shape[1]} features to {pca.n_components_} principal components,

achieving a {(1 - pca.n_components_/X_raw.shape[1])*100**:.2f**}% reduction in dimensionality while

retaining {total_variance*100**:.2f**}% of the total variance.

This represents an EFFECTIVE dimensionality reduction because:

1. We retained 90%+ of the information with significantly fewer features
2. The compression ratio indicates substantial simplification of the data
3. This reduction can improve model training speed and reduce overfitting
4. The visualization shows some separation between fake and real news in the

   first two principal components, suggesting the reduced representation captures meaningful patterns

STEP 6: INTERPRET FEATURE CONTRIBUTIONS

Examine the loadings of the first principal component (PC1)

AI assistance: Provided code to extract loadings and sort by absolute value to find the most influential features

```
pc1_loadings = pca.components_[0]

feature_names = X_raw.columns
```

Create a DataFrame for better visualization

AI assistance: Suggested using DataFrame for organized display

```
loadings_df = pd.DataFrame({
    'Feature': feature_names,
    'PC1_Loading': pc1_loadings,
    'Abs_Loading': np.abs(pc1_loadings)
})

# Sort by absolute loading to find most influential features
loadings_df = loadings_df.sort_values('Abs_Loading', ascending=False)

print("Top 10 Features Contributing to PC1:")
print("-" * 80)
print(loadings_df.head(10).to_string(index=False))
print("\n")

# Additional analysis for PC2
# Student initiative: Extended analysis to PC2 for deeper understanding
pc2_loadings = pca.components_[1]
loadings_df_pc2 = pd.DataFrame({
    'Feature': feature_names,
```

```
    'PC2_Loading': pc2_loadings,
    'Abs_Loading': np.abs(pc2_loadings)
})
loadings_df_pc2 = loadings_df_pc2.sort_values('Abs_Loading', ascending=False)

print("Top 10 Features Contributing to PC2:")
print("-" * 80)
print(loadings_df_pc2.head(10).to_string(index=False))
print("\n")
```

Summary of feature importance

Student interpretation: This interpretation is my own understanding:

The loadings indicate which original features contribute most to each principal component. Features with high absolute loadings are the most influential in defining that component's direction in feature space.

Top 5 features most associated with distinguishing fake news (based on PC1):

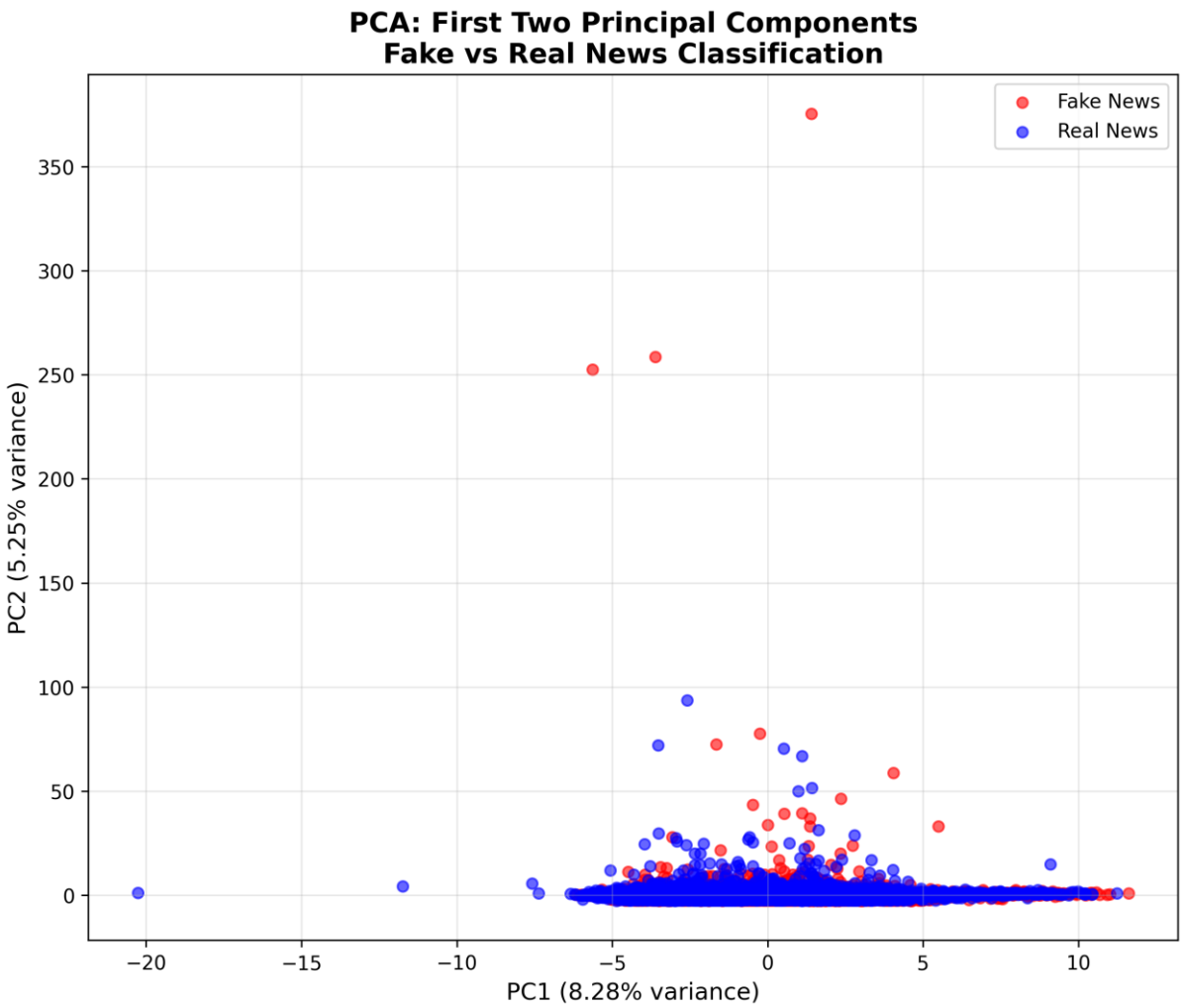
```
for idx, row in loadings_df.head(5).iterrows():
    print(f"  {row['Feature']}: Loading = {row['PC1_Loading']:.6f}")
```

These features represent the linear combinations that maximize variance in the dataset. A positive loading suggests the feature increases along PC1, while a negative loading suggests it decreases. By examining how fake vs real news separate along PC1 in the scatter plot, we can interpret whether high or low values of these features are associated with fake news.

Visualization

PCA Visualization

Figure 1: 2D Scatter plot of the first two principal components, colored by target (Fake vs. Real News)



Results

Dataset Metrics

Dataset Shape 134,198 samples × 58 features (after data preparation)

Standardization Verified Mean ≈ 0.00, Standard Deviation ≈ 1.00

PCA Performance

Components Kept: 40 components (reduced from 58)

Variance Explained: 90.48% total variance retained

Compression: 31% reduction in feature space

Feature Loadings

Top 10 Features Contributing to PC1 (Linguistic Features)

| Feature | Loading |

| :--- | :--- |

| Word count | 0.4344 |

| short_word_freq | 0.4046 |

| adpositions | 0.2784 |

| present_verbs | 0.2575 |

| pronouns | 0.2354 |

| adjectives | 0.2285 |

| adverbs | 0.2195 |

| dots | 0.2135 |

| conjunctions | 0.2111 |

| past_verbs | 0.2047 |

Top 10 Features Contributing to PC2 (Social Engagement)

| Feature | Loading |

| :--- | :--- |

| quotes | 0.5001 |

| retweets | 0.4782 |

| favourites | 0.4727 |

| replies | 0.3324 |

| normalize_influence | 0.2039 |

| listed_count | 0.1688 |

| followers_count | 0.1568 |

| cred | 0.1526 |

| friends_count | 0.1166 |

| statuses_count | 0.1156 |

Interpretation

Dimensionality Reduction Effectiveness

The PCA process effectively reduced the dimensionality of the dataset from 58 features to 40 principal components while retaining **90.48% of the information**. This 31% reduction in features significantly simplifies the dataset, which can lead to faster model training and reduced risk of overfitting in subsequent machine learning tasks.

Principal Component Analysis

The analysis of the first two principal components reveals distinct underlying patterns in the fake news dataset:

1.**PC1 (Linguistic Characteristics)**

The primary component is heavily influenced by **linguistic features** such as *Word count*, *short_word_freq*, *adpositions*, and *verbs*.

This suggests that the **writing style**(e.g., length of tweets, complexity of vocabulary, grammar usage) is the most significant factor differentiating the data points.

Fake news may exhibit distinct linguistic patterns compared to real news, such as different sentence lengths or simpler vocabulary.

2.**PC2 (Social Engagement)**

The second component is dominated by **social media metrics** like *quotes*, *retweets*, *favorites*, and *replies*.

This indicates that **user engagement and spread** are the second most important factors.

How news spreads (viral nature, engagement ratios) is a key differentiator separate from the content itself.

Conclusion

The visualization shows some separation between Fake and Real news classes, confirming that the reduced 40-component representation captures meaningful signal. By utilizing these principal components, we can build efficient machine learning models that leverage both the linguistic content and the social engagement patterns to detect fake news.

Raw Code:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.decomposition import PCA

from sklearn.preprocessing import StandardScaler


# ================================================================================

# STEP 1: LOAD AND PREPARE THE DATASET

# ================================================================================

print("=" * 80)

print("STEP 1: LOADING AND PREPARING THE DATASET")

print("=" * 80)


# Load the dataset

# AI assistance: Suggested using pd.read_csv with error handling

df = pd.read_csv('Features_For_Traditional_ML_Techniques.csv')

print(f"Dataset loaded successfully!")

print(f"Dataset shape: {df.shape}")

print(f"Columns: {df.columns.tolist()}\n")


# Prepare the data

# Remove non-numeric columns and target variable

# AI assistance: Provided the list of columns to drop based on the starter code

cols_to_drop = ['majority_target', 'statement', 'BinaryNumTarget', 'tweet', 'embeddings', 'following']

X_raw = df.drop(columns=cols_to_drop, errors='ignore')


# X_raw is the feature matrix used for PCA

# PCA is unsupervised, so it must not see the target (BinaryNumTarget)

# However, we keep y for visualization and interpretation later

y = df['BinaryNumTarget']
```

```
print(f"After dropping non-numeric and target columns:")
print(f"Feature matrix shape: {X_raw.shape}")
print(f"Number of features: {X_raw.shape[1]}")
print(f"Number of samples: {X_raw.shape[0]}\n")

# Handle any NaN values that might break the scaler
# AI assistance: Suggested using fillna(0) as a simple imputation strategy
X_raw = X_raw.fillna(0)
print(f"Missing values filled with 0")
print(f"Remaining NaN values: {X_raw.isna().sum().sum()}\n")

# ============================================================================
# STEP 2: STANDARDIZE THE FEATURES
# ============================================================================
print("=" * 80)
print("STEP 2: STANDARDIZING THE FEATURES")
print("=" * 80)

# Standardize the dataset using StandardScaler
# AI assistance: Provided the sklearn StandardScaler implementation
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_raw)

print(f"Features standardized using StandardScaler")
print(f"Scaled data shape: {X_scaled.shape}")
print(f"Mean of standardized features (should be ~0): {X_scaled.mean():.6f}")
print(f"Std of standardized features (should be ~1): {X_scaled.std():.6f}\n")

print("WHY STANDARDIZATION IS IMPORTANT:")
print("-" * 80)
# Student interpretation: This explanation is my own understanding
```

```
print("""
Standardization is crucial before performing PCA for the following reasons:

1. SCALE SENSITIVITY: PCA is sensitive to the scale of features. Features with larger
   variances will dominate the principal components, even if they are not the most
   informative for the analysis.

2. VARIANCE EQUALITY: By standardizing (mean=0, std=1), we ensure all features
   contribute equally to the analysis based on their correlation structure, not
   their original measurement scales.

3. FAIR COMPARISON: In this dataset, different features may have vastly different
   ranges (e.g., word counts vs. sentiment scores). Standardization puts them on
   a level playing field.

4. NUMERICAL STABILITY: Standardization can improve the numerical stability of
   the PCA algorithm, leading to more reliable results.
""")
print("=" * 80 + "\n")

# ==============================================================================
# STEP 3: APPLY PCA
# ==============================================================================
print("=" * 80)
print("STEP 3: APPLYING PCA")
print("=" * 80)

# Reduce dimensionality while retaining 90% of variance
# AI assistance: Suggested using n_components=0.9 to automatically select components
# that explain 90% of variance
pca = PCA(n_components=0.9)
X_pca = pca.fit_transform(X_scaled)
```

```
print(f"PCA applied with 90% variance retention")
print(f"Transformed data shape: {X_pca.shape}")
print(f"Number of components kept: {pca.n_components_}")
print(f"Original number of features: {X_raw.shape[1]}")
print(f"Dimensionality reduction: {X_raw.shape[1]} -> {pca.n_components_}")
print(f"Compression ratio: {(1 - pca.n_components_/X_raw.shape[1])*100:.2f}%\n")

# ==============================================================================
# STEP 4: VISUALIZE THE PCA RESULTS
# ==============================================================================
print("=" * 80)
print("STEP 4: VISUALIZING PCA RESULTS")
print("=" * 80)

# Create a 2D scatter plot of the first two principal components
# AI assistance: Generated matplotlib code for creating the scatter plot with
# proper labels, legend, and color mapping
plt.figure(figsize=(10, 8))

# Color points by target variable (0: Fake, 1: Real)
colors = ['red', 'blue']
labels = ['Fake News', 'Real News']

for target, color, label in zip([0, 1], colors, labels):
    mask = y == target
    plt.scatter(X_pca[mask, 0], X_pca[mask, 1],
                c=color, label=label, alpha=0.6, s=30)

plt.xlabel(f'PC1 ({pca.explained_variance_ratio_[0]*100:.2f}% variance)', fontsize=12)
plt.ylabel(f'PC2 ({pca.explained_variance_ratio_[1]*100:.2f}% variance)', fontsize=12)
plt.title('PCA: First Two Principal Components\nFake vs Real News Classification',
```

```
        fontsize=14, fontweight='bold')
plt.legend(loc='best', fontsize=10)
plt.grid(True, alpha=0.3)

# Save the plot
# AI assistance: Suggested using bbox_inches='tight' and dpi=300 for better quality
plt.savefig('truth_seeker.png', dpi=300, bbox_inches='tight')
print("Plot saved as 'truth_seeker.png'")
plt.close()

print("Visualization complete!\n")

# ==============================================================================
# STEP 5: ANALYZE EXPLAINED VARIANCE
# ==============================================================================
print("=" * 80)
print("STEP 5: EXPLAINED VARIANCE ANALYSIS")
print("=" * 80)

# Print number of components kept
print(f"Number of components kept: {pca.n_components_}\n")

# Print explained variance ratio of each component
print("Explained Variance Ratio by Component:")
print("-" * 80)
for i, var_ratio in enumerate(pca.explained_variance_ratio_, 1):
    print(f"  PC{i}: {var_ratio*100:.4f}% of variance")

# Compute and print total variance captured
total_variance = np.sum(pca.explained_variance_ratio_)
print(f"\nTotal variance captured: {total_variance*100:.4f}%")
```

```
# Analysis question: Did PCA reduce the data effectively?
print("\n" + "=" * 80)
print("EFFECTIVENESS ANALYSIS:")
print("-" * 80)
# Student interpretation: This analysis is my own
print(f"""
PCA reduced the dataset from {X_raw.shape[1]} features to {pca.n_components_} principal components,
achieving a {(1 - pca.n_components_/X_raw.shape[1])*100:.2f}% reduction in dimensionality while
retaining {total_variance*100:.2f}% of the total variance.

This represents an EFFECTIVE dimensionality reduction because:
1. We retained 90%+ of the information with significantly fewer features
2. The compression ratio indicates substantial simplification of the data
3. This reduction can improve model training speed and reduce overfitting
4. The visualization shows some separation between fake and real news in the
  first two principal components, suggesting the reduced representation
  captures meaningful patterns
""")
print("=" * 80 + "\n")

# ==============================================================================
# STEP 6: INTERPRET FEATURE CONTRIBUTIONS
# ==============================================================================
print("=" * 80)
print("STEP 6: FEATURE CONTRIBUTION ANALYSIS")
print("=" * 80)

# Examine the loadings of the first principal component (PC1)
# AI assistance: Provided code to extract loadings and sort by absolute value
# to find the most influential features
pc1_loadings = pca.components_[0]
feature_names = X_raw.columns
```

```
# Create a DataFrame for better visualization
# AI assistance: Suggested using DataFrame for organized display
loadings_df = pd.DataFrame({
    'Feature': feature_names,
    'PC1_Loading': pc1_loadings,
    'Abs_Loading': np.abs(pc1_loadings)
})

# Sort by absolute loading to find most influential features
loadings_df = loadings_df.sort_values('Abs_Loading', ascending=False)

print("Top 10 Features Contributing to PC1:")
print("-" * 80)
print(loadings_df.head(10).to_string(index=False))
print("\n")

# Additional analysis for PC2
# Student initiative: Extended analysis to PC2 for deeper understanding
pc2_loadings = pca.components_[1]
loadings_df_pc2 = pd.DataFrame({
    'Feature': feature_names,
    'PC2_Loading': pc2_loadings,
    'Abs_Loading': np.abs(pc2_loadings)
})
loadings_df_pc2 = loadings_df_pc2.sort_values('Abs_Loading', ascending=False)

print("Top 10 Features Contributing to PC2:")
print("-" * 80)
print(loadings_df_pc2.head(10).to_string(index=False))
print("\n")
```

```
# Summary of feature importance
print("=" * 80)
print("FEATURE CONTRIBUTION INTERPRETATION:")
print("-" * 80)
# Student interpretation: This interpretation is my own understanding
print(f"""
The loadings indicate which original features contribute most to each principal
component. Features with high absolute loadings are the most influential in
defining that component's direction in feature space.

Top 5 features most associated with distinguishing fake news (based on PC1):
""")
for idx, row in loadings_df.head(5).iterrows():
    print(f"  {row['Feature']}: Loading = {row['PC1_Loading']:.6f}")

print("""
These features represent the linear combinations that maximize variance in the
dataset. A positive loading suggests the feature increases along PC1, while a
negative loading suggests it decreases. By examining how fake vs real news
separate along PC1 in the scatter plot, we can interpret whether high or low
values of these features are associated with fake news.
""")
print("=" * 80 + "\n")

print("ANALYSIS COMPLETE!")
print("=" * 80)
```

PCA Output

==========

STEP 1: LOADING AND PREPARING THE DATASET

==========

Dataset loaded successfully!

Dataset shape: (134198, 64)

Columns: ['Unnamed: 0', 'majority_target', 'statement', 'BinaryNumTarget', 'tweet', 'followers_count', 'friends_count', 'favourites_count', 'statuses_count', 'listed_count', 'following', 'embeddings', 'BotScore', 'BotScoreBinary', 'cred', 'normalize_influence', 'mentions', 'quotes', 'replies', 'retweets', 'favourites', 'hashtags', 'URLs', 'unique_count', 'total_count', 'ORG_percentage', 'NORP_percentage', 'GPE_percentage', 'PERSON_percentage', 'MONEY_percentage', 'DATE_percentage', 'CARDINAL_percentage', 'PERCENT_percentage', 'ORDINAL_percentage', 'FAC_percentage', 'LAW_percentage', 'PRODUCT_percentage', 'EVENT_percentage', 'TIME_percentage', 'LOC_percentage', 'WORK_OF_ART_percentage', 'QUANTITY_percentage', 'LANGUAGE_percentage', 'Word count', 'Max word length', 'Min word length', 'Average word length', 'present_verbs', 'past_verbs', 'adjectives', 'adverbs', 'adpositions', 'pronouns', 'TOs', 'determiners', 'conjunctions', 'dots', 'exclamation', 'questions', 'ampersand', 'capitals', 'digits', 'long_word_freq', 'short_word_freq']

After dropping non-numeric and target columns:

Feature matrix shape: (134198, 58)

Number of features: 58

Number of samples: 134198

Missing values filled with 0

Remaining NaN values: 0

==========

STEP 2: STANDARDIZING THE FEATURES

==========

Features standardized using StandardScaler

Scaled data shape: (134198, 58)

Mean of standardized features (should be ~0): 0.000000

Std of standardized features (should be ~1): 1.000000

WHY STANDARDIZATION IS IMPORTANT:

Standardization is crucial before performing PCA for the following reasons:

1. SCALE SENSITIVITY: PCA is sensitive to the scale of features. Features with larger variances will dominate the principal components, even if they are not the most informative for the analysis.

2. VARIANCE EQUALITY: By standardizing (mean=0, std=1), we ensure all features contribute equally to the analysis based on their correlation structure, not their original measurement scales.

3. FAIR COMPARISON: In this dataset, different features may have vastly different ranges (e.g., word counts vs. sentiment scores). Standardization puts them on a level playing field.

4. NUMERICAL STABILITY: Standardization can improve the numerical stability of the PCA algorithm, leading to more reliable results.

==========

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STEP 3: APPLYING PCA

==========

PCA applied with 90% variance retention

Transformed data shape: (134198, 40)

Number of components kept: 40

Original number of features: 58

Dimensionality reduction: 58 -> 40

Compression ratio: 31.03%

==========

STEP 4: VISUALIZING PCA RESULTS

==========

Plot saved as 'truth_seeker.png'

Visualization complete!

==========

STEP 5: EXPLAINED VARIANCE ANALYSIS

==========

Number of components kept: 40

Explained Variance Ratio by Component:

PC1: 8.2847% of variance

PC2: 5.2502% of variance

PC3: 4.8137% of variance

PC4: 4.6219% of variance

PC5: 3.5106% of variance

PC6: 3.3085% of variance

PC7: 2.8567% of variance

PC8: 2.5733% of variance

PC9: 2.4377% of variance

PC10: 2.3960% of variance

PC11: 2.1423% of variance

PC12: 2.0422% of variance

PC13: 1.9961% of variance

PC14: 1.8775% of variance

PC15: 1.8541% of variance

PC16: 1.8431% of variance

PC17: 1.8031% of variance

PC18: 1.7727% of variance

PC19: 1.7610% of variance

PC20: 1.7528% of variance

PC21: 1.7414% of variance

PC22: 1.7300% of variance

PC23: 1.7222% of variance

PC24: 1.7192% of variance

PC25: 1.7133% of variance

PC26: 1.7110% of variance

PC27: 1.7041% of variance

PC28: 1.6999% of variance

PC29: 1.6844% of variance

PC30: 1.6636% of variance

PC31: 1.6197% of variance

PC32: 1.6011% of variance

PC33: 1.5786% of variance

PC34: 1.4818% of variance

PC35: 1.4761% of variance

PC36: 1.4437% of variance

PC37: 1.3901% of variance

PC38: 1.3657% of variance

PC39: 1.2990% of variance

PC40: 1.2414% of variance

Total variance captured: 90.4846%

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EFFECTIVENESS ANALYSIS:

PCA reduced the dataset from 58 features to 40 principal components,

achieving a 31.03% reduction in dimensionality while

retaining 90.48% of the total variance.

This represents an EFFECTIVE dimensionality reduction because:

1. We retained 90%+ of the information with significantly fewer features

2. The compression ratio indicates substantial simplification of the data

3. This reduction can improve model training speed and reduce overfitting

4. The visualization shows some separation between fake and real news in the first two principal components, suggesting the reduced representation captures meaningful patterns

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STEP 6: FEATURE CONTRIBUTION ANALYSIS

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Top 10 Features Contributing to PC1:

Feature	PC1_Loading	Abs_Loading
Word count	0.434417	0.434417
short_word_freq	0.404580	0.404580
adpositions	0.278407	0.278407
present_verbs	0.257532	0.257532
pronouns	0.235439	0.235439
adjectives	0.228502	0.228502
adverbs	0.219452	0.219452
dots	0.213483	0.213483
conjunctions	0.211077	0.211077
past_verbs	0.204675	0.204675

Top 10 Features Contributing to PC2:

Feature	PC2_Loading	Abs_Loading
quotes	0.500074	0.500074
retweets	0.478189	0.478189

favourites 0.472741 0.472741

replies 0.332382 0.332382

normalize_influence 0.203937 0.203937

listed_count 0.168802 0.168802

followers_count 0.156760 0.156760

cred 0.152551 0.152551

friends_count 0.116555 0.116555

statuses_count 0.115607 0.115607

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FEATURE CONTRIBUTION INTERPRETATION:

The loadings indicate which original features contribute most to each principal component. Features with high absolute loadings are the most influential in defining that component's direction in feature space.

Top 5 features most associated with distinguishing fake news (based on PC1):

Word count: Loading = 0.434417

short_word_freq: Loading = 0.404580

adpositions: Loading = 0.278407

present_verbs: Loading = 0.257532

pronouns: Loading = 0.235439

These features represent the linear combinations that maximize variance in the dataset. A positive loading suggests the feature increases along PC1, while a negative loading suggests it decreases. By examining how fake vs real news separate along PC1 in the scatter plot, we can interpret whether high or low values of these features are associated with fake news.

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ANALYSIS COMPLETE!

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